Quality Vision Gap Analysis

Easy Explanation Notes for Manager Discussion

Use this to understand the source links and explain the gap analysis in simple language.

Simple idea: We are not guessing. We are checking what manufacturers are using and what competitors are offering. Then we compare that with our approved prototype and say what is missing or not clearly visible.

1. The Simple Logic of the Document

- **Step 1:** Read the public source link.
- **Step 2:** Note only what the source actually says they use or offer.
- **Step 3:** Compare that with our approved Quality Vision prototype screenshots.
- **Step 4:** Write the gap only if our prototype does not clearly show that capability.
- **Step 5:** Explain why adding it will make the prototype stronger for client demos.

Important caution: Do not say “this customer needs this” unless the link says it. We should say “this source shows this capability is already being used/offered, so our prototype should consider a similar capability.”

2. Main Gaps in Very Simple Words

Gap	Why we got this gap	Simple explanation
Class-wise model performance	Sources show inspection of specific defect types/scenarios.	Overall model score is not enough. We should show how well the model works for each defect type.
Quality Vision AI Assistant / inspection recommendations	BMW source shows AI giving tailored inspection recommendations.	The dashboard should not only show data; it should also help users decide what to inspect or act on.
Root-cause analysis	Bosch source shows AI finding causes of quality problems.	Users need to know why defects are happening, not only how many defects happened.
Workflow / NCR / SAP or MES action	SAP and Wipro sources show inspection inside workflow and reporting.	After a defect is detected, the user should be able to assign, create NCR, sync, and close it.
Executive / business impact view	Wipro talks about ROI, cost, labor reduction and business impact.	Clients will ask how this saves money and improves operations.
MLOps / model governance	Mu Sigma source talks about MLOps and production-grade AI.	For real deployment, model version, monitoring, retraining and rollback should be clear.

3. Source-wise Explanation in Simple Language

3.1 Huawei: Foxconn / Powerleader / Midea

Link: https://e.huawei.com/en/case-studies/industries/manufacturing/ai-quality-inspection-2023

Use Ctrl+F for: Foxconn partnered with Huawei; silicone grease; nameplates; Powerleader; Midea Group; adjustable leg inspection; above 99%

- **What the source says:** Huawei says Foxconn uses AI inspection for smart PV controllers. It checks silicone grease color/quantity and nameplates. It also says Powerleader used AI for incoming quality control, process quality check and packaging inspection. It says Midea used AI inspection in refrigerator factories for adjustable leg, ecolabel, brand logo and condenser application inspection.

- **Gap we wrote:** This is why we wrote “Class-wise Model Performance”. The source shows many defect/use-case scenarios. Our approved prototype shows overall ML metrics, but not clear per-defect reliability.

- **How to explain to manager:** “Huawei’s example is not generic quality inspection. It lists specific checks. So in our prototype, we should show how the model performs for each defect type or inspection scenario.”

Careful wording: Do not say Huawei shows a class-wise dashboard. The link shows specific inspection scenarios and accuracy; our gap is derived from that.

3.2 BMW: GenAI4Q

Link: https://www.press.bmwgroup.com/global/article/detail/T0449729EN/artificial-intelligence-as-a-quality-booster?language=en

Use Ctrl+F for: GenAI4Q; tailored inspection recommendations; 1,400 vehicles

- **What the source says:** BMW says its GenAI4Q pilot gives tailored inspection recommendations for around 1,400 vehicles manufactured each day.

- **Gap we wrote:** This is why we wrote “Quality Vision AI Assistant / Inspection Recommendations”. Our approved prototype shows dashboards and insights, but not a strong quality-specific assistant that recommends inspection actions.

- **How to explain to manager:** “BMW uses AI to recommend what to inspect. So our Quality Vision should also guide the user, not only display numbers.”

Careful wording: Do not say BMW uses our Quality Vision Agent. Say BMW uses AI for quality inspection recommendations.

3.3 Bosch: AI Analytics Platform

Link: https://www.bosch.com/stories/ai-in-manufacturing/

Use Ctrl+F for: AI Analytics Platform; 1,400 production lines; causes of quality problems; trace errors

- **What the source says:** Bosch says its AI Analytics Platform is used on over 1,400 production lines. It analyzes manufacturing data and finds causes of quality problems. It also mentions cameras and dashboards that help trace errors.

- **Gap we wrote:** This is why we wrote “Root-cause analysis” and “stronger filters”. Our prototype shows line quality and trends, but not strong why-this-happened analysis.

- **How to explain to manager:** “Bosch is using AI to find causes of quality problems. Our prototype should also help quality teams understand root cause by plant, line, shift and component.”

Careful wording: Do not say Bosch uses the same UI. Say Bosch publicly describes AI analytics for quality-problem causes at production-line scale.

3.4 Wipro: InspectAI

Link: https://www.wipro.com/ai/inspectai/

Use Ctrl+F for: automates visual inspection; Streamline Inspection Workflow; Business Impact; Maximize ROI; Reduce labor and operational costs

• **What the source says:** Wipro says InspectAI automates visual inspection with AI and computer vision. It also mentions workflow, assignment, reporting, compliance, ROI, labor reduction and cost reduction.

• **Gap we wrote:** This is why we wrote “Executive Cockpit”, “Assign/Escalate/NCR workflow” and “Export/Reports”. Our prototype is strong technically, but business value and workflow actions should be more visible.

• **How to explain to manager:** “Wipro is presenting visual inspection as a business and workflow solution, not only an AI screen. So our prototype should clearly show ROI, workflow and reporting.”

Careful wording: Do not say Wipro is identical to our prototype. Say it is a competitor benchmark for AI visual inspection and inspection management.

3.5 LandingAI: Manufacturing computer vision

Link: https://landing.ai/industries/manufacturing

Use Ctrl+F for: Good / No Good; Part Missing; Scratch; Smudge; Quality Control Inspection

• **What the source says:** LandingAI says manufacturers can use computer vision to classify parts as Good/No Good or identify specific issues such as Part Missing, Scratch and Smudge.

• **Gap we wrote:** This supports “defect class management” and “class-wise model performance”. Our prototype shows defect types, but not model reliability by each defect class.

• **How to explain to manager:** “LandingAI talks in terms of specific defect categories. So we should also show the AI’s reliability for each category.”

Careful wording: Do not say LandingAI gives our exact metrics. Use it as market evidence for defect-specific inspection.

3.6 SAP: Digital Manufacturing visual inspection

Link: https://www.sap.com/use-cases/identify-nonconformance-early-in-the-manufacturing-process

Use Ctrl+F for: AI-driven visual inspections; shape, assembly, or surface; annotate results; digital workflow

• **What the source says:** SAP says Digital Manufacturing supports manual and AI-driven visual inspections to identify flaws in shape, assembly or surface. It also mentions annotation, tasks, dashboards and digital workflow.

• **Gap we wrote:** This is why we wrote “SAP QM/MES action”, “NCR workflow” and “status tracking”. Our prototype mentions SAP QM, but does not clearly show the full lifecycle after defect detection.

• **How to explain to manager:** “SAP shows inspection as part of a workflow. In our prototype, after detecting a defect, the user should see actions like create NCR, sync SAP QM, assign and close.”

Careful wording: Do not say SAP is a customer. Say SAP is a third-party benchmark for enterprise visual inspection workflow.

3.7 Mu Sigma: Analytics competitor / adjacent evidence

Link: https://www.mu-sigma.com/partners/microsoft/

Use Ctrl+F for: MuNLQ; Digital Twin; Production Yield; MLOps; Interactive Reporting

• **What the source says:** Mu Sigma lists AI/ML, MLOps, MuNLQ, Digital Twin, production yield and interactive reporting. MuNLQ is a natural-language query capability for business users.

• **Gap we wrote:** This supports “AI Assistant”, “MLOps/model governance” and “executive reporting”. It is adjacent evidence, not direct Quality Vision proof.

• **How to explain to manager:** “Mu Sigma is not showing a direct Quality Vision Agent here, but it shows competitor-level analytics capabilities like NLQ, MLOps and reporting. Our prototype should not look weaker in these areas.”

Careful wording: Do not say Mu Sigma has a Quality Vision Agent unless we find a direct public source.

4. Example You Asked About: Foxconn / Powerleader / Midea

Question: How did we get “Class-wise Model Performance and defect-specific evidence” from the Huawei source?

Simple answer: Because Huawei lists specific inspection scenarios, not just a generic AI inspection. It talks about silicone grease, nameplates, adjustable leg, ecolabel, logo and condenser checks. If a solution is used for many different defect scenarios, then a client may ask how reliable the model is for each scenario. Our approved prototype shows overall mAP/precision, but not per-defect performance. So our gap is class-wise model performance.

Meeting line: “This source lists specific AI inspection checks. So our prototype should not only show overall model accuracy. It should show how the model performs for each defect type, because each defect has different business risk.”

5. Quick Answers for Manager Questions

• **Where did this insight come from?:** From the public source link. I used Ctrl+F keywords to verify the exact statement.

• **Are these companies our customers?:** No. They are public manufacturing/competitor references only.

• **Are we saying the source has the same feature as our prototype?:** No. We are saying the source shows a capability being used/offered, and our prototype should be checked against it.

• **Why class-wise model performance?:** Because sources mention specific defect scenarios/classes. Overall accuracy is not enough for trust.

• **Why workflow/SAP/NCR?:** Because visual inspection should lead to action: assign, create NCR, sync, report and close.

• **Why executive cockpit?:** Because competitors present business impact such as ROI, labor saving, cost reduction and production quality.

6. Final Summary to Say in the Meeting

“The approved prototype is already strong as a Quality Vision dashboard. But public sources show that manufacturers and competitors are using or offering AI visual inspection with defect-specific checks, recommendations, workflows, reporting, business impact and governed AI. So our gaps are not random. They are the missing product-readiness items needed to make the prototype stronger for client discussions.”